

```

=== Step 6: GetAllAssets ===
[
  {
    "AppraisedValue": 10000,
    "Color": "blue",
    "ID": "Asset-2",
    "Owner": "Arun",
    "Size": 5
  },
  {
    "AppraisedValue": 300,
    "Color": "blue",
    "ID": "asset1",
    "Owner": "Tomoko",
    "Size": 5,
    "docType": "asset"
  },
  {
    "AppraisedValue": 400,
    "Color": "red",
    "ID": "asset2",
    "Owner": "Brad",
    "Size": 5,
    "docType": "asset"
  }
]

```

Figure 5: Full ledger state; Asset-2 shows Arun as current owner

without breaking every hash that follows it.

Now that we have gone through the main experiment, here are a few more things you can try with the same network and chaincode still running.

Experiment 3 — Querying asset history: The asset-transfer-basic chaincode has a `GetAssetHistory` function. Unlike `ReadAsset`, which only shows you the current state, this one pulls up every recorded state of an asset from the beginning. For Asset-2, it will show two entries —when Suyash first created it and when it was transferred to Arun, each with a timestamp and transaction ID. Nothing can be removed or reordered. This is exactly what makes blockchain useful for audits and proving a chain of custody.

Experiment 4 — Updating an asset: `UpdateAsset` lets you change any field of an existing asset — its colour, size, or appraised value — without touching the ID or owner. You could try changing the appraised value of Asset-2 from 10000 to 15000 and then querying it back to confirm. The old value does not disappear; it stays in the history. What this shows is that even a simple edit goes through the full endorsement-ordering-commit cycle, same as a transfer. Nothing skips the queue.

Experiment 5 — Rich query using

CouchDB: Because the test network runs CouchDB as the state database, you can query assets by their field values instead of just by ID. This is something you cannot do with the default LevelDB. For example, you could search for all assets where Owner equals Arun, or all assets with an appraised value above a certain number. In a real deployment, where the ledger has thousands of assets, this kind of query becomes very useful when a participant wants to see everything they own without needing to know each ID upfront.

To shut the network down, use the command:

```
./network.sh down
```

This stops all containers and removes the generated crypto material. Running `network.sh up` again regenerates everything from scratch.

This experiment covered the full lifecycle of a digital asset on a private blockchain network—setup, chaincode deployment, asset creation, ownership transfer, and

ledger audit—running entirely on a PC with Docker and WSL 2. The same Hyperledger Fabric framework, with the same endorsement-ordering-commit architecture, runs in production supply chains, trade finance systems, and healthcare data networks.

The test network makes it possible to understand exactly how those systems work without needing any enterprise infrastructure. The next step is to write custom chaincode and define your own data model and business rules, or connect a web application to the ledger using the Fabric Gateway SDK. **END** 🐧

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